

- 17 D** Since temperature is constant, $\Delta U = 0$.
 Since the kettle is rated as 500 W, rate of electrical work done on the coil is 500 W.
 The heating coil continues to lose heat to the water. From the first law of thermodynamics,

$$\frac{dQ}{dt} = -500 \text{ W}.$$

- 18 A** Taking towards the right as positive